

Neural Differential Utility: Accepted Adaptive Service Control and Certified Value-Gradient Learning

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Abstract. Can an externally priced service protocol adapt without reducing customer utility or service? We study persistent tiers, physical stock, costly reconfiguration, and participation at every review. A full-tree balance characterization relates accepted adaptation to continuation prices, boundary scarcity, and switching capacities. On scalar trees, exact payment-transfer coordinates yield a sparse critical-friction program, a two-pass dual repair, and a constructive profitable policy when friction is below a separating threshold. A computable policy-loss bound penalizes inaccurate prices and switching faces without requiring the unknown optimal face. For quadratic resources, verified parameter cells support compatible scalar value experts and neural routing with explicit rejection and refinement. A model-derived validation range gives simultaneous expected-gain and regret guarantees for frozen complete pipelines. Nonzero-switching experiments exercise the structural theory, while matched-accuracy comparisons include classical cell lookup, exact audits, offline construction, and actual fallback. Independent validation establishes positive expected gain over static in the declared synthetic distribution, not universal neural superiority. Earlier information-value results and unfavorable nonlinear transfer evidence are retained. The results connect service acceptance, implementable learning, and policy certification without conflating these guarantees.

Keywords: accepted service control; continuation participation; value-gradient learning.

Subject classifications: Inventory/production: service contracts; Dynamic programming/optimal control: stochastic control; Analysis of algorithms: approximation guarantees.

Area of review: Stochastic Models.

1 Introduction

A service provider can revise both physical capacity and the service tier attached to it. A higher tier earns a premium but can also increase indemnity and maintenance costs; replacing an installed tier consumes resources. Adaptation is useful only when these gains survive the customer’s commitments and comparison with a properly optimized outside protocol. We ask when an accepted adaptive protocol creates value, how switching friction changes that answer, and how a learned implementation can be evaluated under the same acceptance requirements.

We study Neural Differential Utility as joint control of physical activity and a persistent, externally priced contractual state. Regimes, stock, tiers, and realized fulfillment are observable and contractible. The tariff and accounting rule are fixed before optimization. The provider commits to an announced contingent protocol; in the continuation model the customer can leave at every review. There is no private type or hidden effort. The model is an ordinary Markov decision process with expanded state and action spaces, not a claimed escape from stochastic control. Its distinctive comparison preserves the customer’s utility, expected fill, and physical resource commitments relative to an optimized static arrangement.

Our first contribution identifies the structure of accepted adaptation on a full scenario tree. With absolute switching costs, marginal operating rewards must be balanced by continuation prices, boundary scarcity, and bounded switching tensions. On a scalar tree, eliminating the tensions yields cumulative subtree cuts and an exact critical-friction program. This extends the earlier local edge criterion beyond a two-review adjustment. The general balance theorem admits vector tiers, joint smooth graph costs, multiple commitments, signed net-payment rates, boundary solutions, and nonconstant outside protocols. With all temporal and installation edges penalized, a constant protocol with no initial mismatch becomes optimal at sufficiently large friction. For an already-changing outside protocol, the optimality region can instead have an upper endpoint. We give both a complete characterization and a constructive improving policy when the balance fails. The price–capacity margin bounds accepted value; exact equality requires the stated quadratic curvature and feasible-step clearance, rather than holding for arbitrary boundary problems.

Our second contribution makes the structure algorithmic. On a scalar tree, nonnegative payment transfers parameterize the entire accepted polytope. A two-pass tension repair gives an explicit upper bound on critical friction, while a separating transfer provides both a lower bound and a profitable accepted decision. Keeping the incidence factors separate prevents quadratic fill-in on high-branching trees. Sparse linear programming and a bracketed first-order method are evaluated separately; linear work per iteration does not imply a linear-time optimization algorithm.

Our third contribution supplies an implementable connection to learning. An approximate-face certificate uses only a candidate and predicted prices; wrong switching signs and slack commitments incur explicit loss penalties. Under quadratic resources, offline price maps define verifiable parameter cells and compatible scalar value experts. A neural router proposes cells, but feasibility, price signs, and an exact policy audit determine acceptance. This removes the unknown query-time face from the sufficient condition. Rejection and actual refinement remain part of the complete algorithm, and classical lookup of the same cells remains a comparator.

Our fourth contribution executes the theory and validation. Nonzero-switching experiments trace ac-

cepted decisions through critical friction. Independent samples evaluate frozen neural and nearest-anchor pipelines, with a model-derived range and a positive expected-gain bound relative to static. Matched-accuracy tests include the entire deployment and offline cost. The exact canonical information hierarchy retains its total accepted improvement of 0.63666527 and inherited-tier contribution of 0.01929038. The earlier non-centered boundary experiments and severe neural transfer failures remain intact in the companion. No field calibration, general distribution-shift guarantee, or universal gradient-supervision advantage is inferred.

1.1 Related literature and the incremental contribution

Inventory decisions and service contracts have long been studied jointly. Henig et al. (1997) embed inventory decisions in supply-contract design; Liang and Atkins (2013); Sieke et al. (2012); Katok et al. (2008) study agreements, penalties, coordination, and review periods. Dynamic stocking under service-level agreements is established (Hosseini et al., 2022; Caggiano et al., 2007; Abbasi et al., 2022). Here the static-contract comparator can already adapt physical stock to the regime. The additional decisions are accepted revisions of contractual tiers, not state-aware inventory actions withheld from the benchmark.

Private information and unobserved effort create different constraints. Nasser and Turcic (2019) study temporary contract adjustment with a private demand forecast; Plambeck and Zenios (2003) study incentive-efficient control with unobserved effort. We impose public-state continuation participation, not truth telling or hidden-action incentive compatibility. Combined pricing and inventory optimization is also established (Federgruen and Heching, 1999); our tariff and demand law remain external to the provider’s control.

Constrained Markov decision processes, occupation measures, and Lagrangian resource prices are classical (Altman, 1999). Conditional promised utility supports recursive contracting (Spear and Srivastava, 1987), while self-enforcing wage contracts analyze participation over time (Thomas and Worrall, 1988). Our remaining-payment state is a service-control specialization, not a new principle of dynamic programming. The contribution is the exact continuation-price and switching-capacity characterization, including its cumulative cut form and friction comparative statics, under the stated observable-service institution.

Monotone selections follow Topkis (1978), and hysteretic control has predecessors including Hipp and Holzbaur (1988). Supporting-price stock identification, normal cones, Bregman decompositions, and strong-convexity residual bounds use standard convex tools (Boyd and Vandenberghe, 2004). We state those tools explicitly so that their specialization is not mistaken for independent mathematical novelty. Approximate dynamic programming also uses optimization to connect approximate values and decisions (de Farias and Van Roy, 2003). Our complete-tree audit retains all participation rows and boundary prices rather than relying on a collection of unverified one-step errors.

Derivative-weighted fitting follows Sobolev training (Czarnecki et al., 2017); graph filtering follows Defferrard et al. (2016). Safe and constrained learning studies cumulative constraints and, in some models, unknown safe regions (Satija et al., 2020; Wachi and Sui, 2020). Here the feasible polyhedron is known; learning proposes a policy and a separate calculation certifies it. The acceptance-face theorem and certificate-risk selection result explain the role and limits of this separation. We benchmark SciPy’s sequential quadratic programming implementation alongside the preserved sparse Newton methods (Virtanen et al., 2020); no custom implementation is declared the practical optimization frontier. Forward–backward value-process

methods (Ma et al., 1994; Han et al., 2018) and Markov-chain approximation (Kushner and Dupuis, 2001) support the separately preserved continuous-time material, not the finite-tree theorem by analogy.

Graph total variation and regularization paths have established dual and algorithmic representations (Tibshirani and Taylor, 2011; Arnold and Tibshirani, 2016). Explicit constrained control uses multiparametric quadratic programs (Bemporad et al., 2002), and safe screening uses gap-based regions (Ferrocoq et al., 2015). Our contribution is not to rename those methods: it is the continuation-transfer representation, constructive price repair, and their integration with verified scalar experts and independently validated accepted deployment.

The main paper contains the institution, essential proofs, canonical comparison, and current implementation results. Optional predecessor proofs and diagnostic tables are in the electronic companion. The historical supplement preserves the full earlier diffusion, continuous-time, and computational development without making it a prerequisite for understanding the current results.

2 An externally priced inventory contract

There are periods $t = 0, \dots, T - 1$. The observed state is (z, q) , where z is a demand regime in a finite ordered set $\mathcal{Z}$ and $q \in [0, 1]$ is the inherited contract. Conditional demand D_z is nonnegative with finite first moment. A decision consists of stock $S \in \mathcal{S} \subset [0, \bar{S}]$ and contract $\theta \in \Theta \subset [0, 1]$. Both feasible sets are fixed compact chains; finite grids are allowed. Unused stock perishes and unmet demand is lost. Thus no inventory or backlog is carried. The next regime has exogenous transition matrix P , and the next inherited contract equals the chosen contract.

Write the expected overage and shortage as

$$O_z(S) = \mathbb{E}(S - D_z)^+, \quad L_z(S) = \mathbb{E}(D_z - S)^+, \quad C_z(S) = cS + hO_z(S) + pL_z(S). \quad (1)$$

Here $c, h, p \geq 0$ are externally fixed physical-cost coefficients. The provider receives premium $r_z\theta$, pays incremental liability $\kappa\theta L_z(S)$, pays maintenance $m\theta^2$, and pays adjustment $\lambda(\theta - q)^2$, where $\kappa, m, \lambda > 0$. Expected one-period net reward is

$$g(z, q, S, \theta) = r_z\theta - C_z(S) - \kappa\theta L_z(S) - m\theta^2 - \lambda(\theta - q)^2. \quad (2)$$

The expectation in this expression is over demand, not over the decision. Premium rates, liability coefficients, and physical-cost weights cannot be controlled. The contract is a real cash-flow commitment. Its level affects both reward and the future state; it is not an estimator of a preference parameter.

2.1 A customer-accepted master agreement

The contract is a publicly specified menu, not a penalty that the provider can change after performance is observed. Before period zero, a risk-neutral customer and provider sign a master agreement. It specifies the premium schedule $r_z\theta$, the indemnity $\kappa\theta(D_z - S)^+$, the horizon, and a contingent operating protocol. Regime, tier, stock, demand, and fulfillment are observable and contractible. At each review the provider implements

the announced protocol after observing (z, q) and before demand is realized. Public randomization is permitted when specified in the agreement. The customer commits to the contingent menu, not to a new purchase at each review. Transfers and compliance with the protocol are enforceable. No hidden effort or private type is assumed.

Let $f_z(S) = \mathbb{E}D_z - L_z(S)$ be expected units served. For a fixed service benefit $\nu > 0$, the customer's incremental one-period utility is

$$u(z, S, \theta) = \nu f_z(S) + \kappa \theta L_z(S) - r_z \theta. \quad (3)$$

Indemnity is a transfer from the provider to this customer. Maintenance and adjustment are real resource expenditures, not payments to the customer. Write $F^\pi = \mathbb{E}^\pi \sum_t \beta^t f_{z_t}(S_t)$, $D = \mathbb{E} \sum_t \beta^t \mathbb{E}D_{z_t}$, and $U^\pi = \mathbb{E}^\pi \sum_t \beta^t u(z_t, S_t, \theta_t)$. Demand is exogenous, so D is independent of the protocol. The agreement is accepted only if

$$U^\pi \geq \underline{U}, \quad F^\pi \geq \gamma D. \quad (4)$$

These are ex ante participation and expected discounted fill commitments. They are not period-by-period chance constraints or a right to renegotiate after each regime realization. A customer with such rights would require continuation participation constraints and an additional promised-utility state.

A fixed retainer A_0 can fund provider participation. Total provider profit is $A_0 + W^\pi$ and total customer surplus is $U^\pi - A_0$; the retainer does not affect policy comparisons. In the numerical study $A_0 = 6$, $\nu = 2$, and the customer's outside alternative is the best fixed-contract arrangement, including the same retainer. Both that arrangement and the accepted adaptive arrangement satisfy a zero provider outside option. We report incremental W and U so that the historical accounting remains exactly comparable.

The institution fixes the menu and designs the contingent protocol subject to acceptance. It does not claim to solve private-information screening or to endogenize every price in the menu. The unconstrained Bellman equation below is the provider relaxation and is also the problem when the acceptance constraints are slack. Section EC.1 solves the binding-acceptance problem. A lower fill rate or customer surplus in the relaxation is not presented as an improvement under the accepted agreement.

For discount $\beta \in (0, 1]$ and zero terminal reward, the Bellman equation is

$$V_t(z, q) = \max_{S, \theta} \left\{ g(z, q, S, \theta) + \beta \sum_{z'} P_{zz'} V_{t+1}(z', \theta) \right\}, \quad V_T = 0. \quad (5)$$

Finite actions make existence immediate. For compact intervals, bounded demand moments, continuity of the stage reward, and induction give a continuous value and an attained maximum. We select the greatest optimal pair when the lattice conditions below hold.

The exact expanded-action representation and the provider-relaxation structure are retained in EC Section EC.4.1. Acceptance is imposed in the following sections.

3 Identification and a unified accepted-control problem

This section supplies a common formulation for the scalar contract model, joint service portfolios, and continuation participation. Critical-fractile analysis and convex first-order optimality are established tools; the supporting-price interpretation identifies their role here. The full-tree cut characterization in Section 4 supplies the switching and continuation structure; the earlier scalar edge criterion is retained in Appendix A.

3.1 Demand primitives and identified physical decisions

Let D_z be nonnegative demand with finite mean, on an exogenous regime process whose occupation probabilities do not depend on the policy. On a compact stock interval $I_z = [l_z, u_z]$, $l_z < u_z$, write

$$C_z(S) = cS + h\mathbb{E}(S - D_z)^+ + p\mathbb{E}(D_z - S)^+, \quad f_z(S) = \mathbb{E} \min(S, D_z), \quad \Psi_z(S) = C_z(S) - \xi f_z(S).$$

Here $\xi > 0$ is a supporting service price, not a contractual transfer chosen by the provider. Denote the demand distribution function by H_z , avoiding confusion between physical fill and resource cost. The critical level is $\tau = (p + \xi - c)/(h + p + \xi)$, with $h + p + \xi > 0$.

Proposition 3.1 (Identified stocks, boundaries, and slack commitments). *Suppose s_z uniquely minimizes Ψ_z on I_z . Then the commitments $F^\pi \geq F^s$ and $C^\pi \leq C^s$ force $S_t = s_{z_t}$ almost surely at every positive-probability review, and both commitments bind. This statement includes randomized policies.*

For an interior minimizer, the one-sided derivatives satisfy

$$\Psi'_{z,-}(S) = c - p - \xi + (h + p + \xi)H_z(S-), \quad \Psi'_{z,+}(S) = c - p - \xi + (h + p + \xi)H_z(S).$$

Thus $H_z(s_z-) \leq \tau \leq H_z(s_z)$. At l_z the condition is $\Psi'_{z,+}(l_z) \geq 0$, and at u_z it is $\Psi'_{z,-}(u_z) \leq 0$. A strictly straddling atom or a continuously increasing crossing gives uniqueness; when no crossing lies in the feasible interval the constrained minimizer can instead be an endpoint. The condition $0 < \tau < 1$ alone does not guarantee an interior feasible quantile.

If, separately, $\Psi_z(S) - \Psi_z(s_z) \geq \alpha|S - s_z|^2$ on every I_z , then relaxed commitments imply

$$\mathbb{E}^\pi \sum_t \beta^t |S_t - s_{z_t}|^2 \leq (\delta_C + \xi \delta_F)/\alpha, \quad F^\pi \geq F^s - \delta_F, \quad C^\pi \leq C^s + \delta_C. \quad (6)$$

A sufficient smooth-demand condition is a density at least $\rho > 0$ throughout the entire feasible stock interval, with $\alpha = (h + p + \xi)\rho/2$.

Proof. The displayed derivatives follow by differentiating expected excess and shortage from the left and right. Convexity gives the endpoint and interior conditions. Sum the nonnegative gaps $\Psi_{z_t}(S_t) - \Psi_{z_t}(s_{z_t})$ under the policy's discounted occupation probabilities. Exogeneity makes the comparator use those same probabilities, while acceptance bounds their sum above by $(C^\pi - C^s) - \xi(F^\pi - F^s) \leq 0$. Uniqueness therefore identifies stock at each reachable review. Under quadratic growth the same sum is at least the left side of (6) times α . The density bound gives $\Psi''_z \geq 2\alpha$ almost everywhere; integration and the constrained

first-order condition prove the growth inequality even at an endpoint. The smooth-growth conclusion is not attributed to the atomic canonical instance. $\square$

For d stocks with arbitrary correlated demands, a concrete vector extension is $C_z(S) = \sum_j C_{zj}(S_j)$ and $f_{zj}(S_j) = \mathbb{E} \min(S_j, D_{zj})$, with a separate commitment $F_j^\pi \geq F_j^s$ for every service and a single total physical-cost cap. Choose $\xi_j > 0$ and the unique constrained marginal quantiles at levels $(p_j + \xi_j - c_j)/(h_j + p_j + \xi_j)$. On a product stock set, $\Psi_z = C_z - \sum_j \xi_j f_{zj}$ has the unique vector minimizer. Correlation is immaterial because these costs are additive expectations. Under coordinate density bounds, (6) holds with a squared Euclidean norm, numerator $\delta_C + \sum_j \xi_j \delta_{F,j}$, and the smallest coordinate curvature. Coupled stock constraints require these coordinate minimizers to be jointly feasible; the statement is not extended to an arbitrary vector service functional by assertion.

3.2 Joint resources and explicit information restrictions

On a finite exogenous tree, let $x_n \in [0, 1]^d$ be the tier vector at decision node n , $w_n > 0$ its discounted probability, and $a_n \in \mathbb{R}^d$ its net payment rates. The fixed initial tier is q_0 . Identified physical cost C^s is constant. Set

$$\mathcal{R}(x) = \sum_n w_n \{M_n(x_n) + A_n(x_n - x_{\text{pa}(n)})\} + \mathcal{G}(x), \quad (7)$$

$$\begin{aligned} W(x) &= p^\top x - \mathcal{R}(x) - C^s, & p_n &= w_n a_n, \\ X &= \{x \in [0, 1]^{Nd} : Bx \leq b, Ex = e\}. \end{aligned} \quad (8)$$

All resource functions are differentiable and convex on a neighborhood of the box. The joint function $\mathcal{G}$ includes, for example, $\sum_n w_n \zeta_n \sum_{(i,j) \in \mathcal{E}_n} v_{nij} (x_{ni} - x_{nj})^2$. This is genuinely nonseparable within a portfolio. It is not represented by a sum of univariate functions. The equations $Ex = e$ identify actions sharing the same permitted information; equality multipliers remain unrestricted, rather than being disguised as payment prices. Each continuation row of B uses discounted subtree payments. Multiplication by a positive node probability converts a conditional participation inequality to this unconditional form.

Let $\bar{x}$ embed the reoptimized static comparator, with commitments defined from that comparator and $\bar{x} \in X$. Put $g = p - \nabla \mathcal{R}(\bar{x})$. Define

$$T_X(\bar{x}) = \overline{\{s(y - \bar{x}) : s \geq 0, y \in X\}}, \quad N_X(\bar{x}) = \{v : v^\top (y - \bar{x}) \leq 0 \forall y \in X\}.$$

Lemma 3.2 (Standard first-order alternative specialized to acceptance). *An accepted policy strictly improves $W(\bar{x})$ if and only if $g \notin N_X(\bar{x})$. Equivalently, the following explicit linear program has a positive value:*

$$\max_d g^\top d \quad \text{s.t.} \quad B_i d \leq 0 \ (B_i \bar{x} = b_i), \quad Ed = 0, \quad \|d\|_\infty \leq 1, \quad d_j \geq 0 \ (\bar{x}_j = 0), \quad d_j \leq 0 \ (\bar{x}_j = 1).$$

For such a direction, let $s_{\max} = \sup\{s \geq 0 : \bar{x} + ud \in X \text{ for } 0 \leq u \leq s\}$. If the directional second derivative

of $\mathcal{R}$ is at most L_d on this segment, then

$$W(\bar{x} + sd) - W(\bar{x}) \geq sg^\top d - L_d s^2/2. \quad (9)$$

For $L_d > 0$, take $s = \min\{s_{\max}, g^\top d/L_d\}$.

Proof. Concavity gives $W(y) - W(\bar{x}) \leq g^\top (y - \bar{x})$. Conversely every direction in the displayed polyhedral tangent cone is feasible on a short segment, and a positive derivative gives improvement. Tangent–normal polarity gives the equivalence. Twice integrating the directional curvature bound proves (9). This is the ordinary first-order test and curvature estimate for convex optimization, not a new optimality principle (Boyd and Vandenberghe, 2004). $\square$

Lemma 3.3 (Resource loss and acceptance slack). *At an optimizer x^* , suppose $\eta \geq 0$, $\xi \in \mathbb{R}^{\text{rows}(E)}$, and $v \in N_{[0,1]^{Nd}}(x^*)$ satisfy $p - \nabla \mathcal{R}(x^*) = B^\top \eta + E^\top \xi + v$ and $\eta^\top (b - Bx^*) = 0$. For every $x \in X$,*

$$W^* - W(x) = D_{\mathcal{R}}(x, x^*) + \eta^\top (b - Bx) + v^\top (x^* - x),$$

where $D_{\mathcal{R}}(x, x^*) = \mathcal{R}(x) - \mathcal{R}(x^*) - \nabla \mathcal{R}(x^*)^\top (x - x^*)$. Every term is nonnegative.

Proof. Add and subtract the linearization of $\mathcal{R}$, substitute stationarity, and use complementary slackness and $E(x - x^*) = 0$. The signs follow from convexity, feasibility, and the outward-normal definition. For a differentiable convex objective over a nonempty polyhedron, first-order optimality and the polyhedral normal-cone representation provide these multipliers; no interior Slater point is required for that representation. This standard KKT decomposition (Boyd and Vandenberghe, 2004) separates real resource inefficiency from unused customer commitments in the present model. It is not claimed as a novel algebraic identity. $\square$

4 Full-tree switching, continuation prices, and critical friction

The scalar edge test in Appendix A becomes a capacitated balance condition when changing a tier has an absolute cost. The balance formulation also accommodates nonconstant outside protocols, boundary tiers, several commitments, and vector services. It does not divide by a tariff and therefore remains meaningful for zero or negative net payments. Negative net payments can arise when expected indemnity exceeds the premium; they need not be interpreted as negative gross prices.

4.1 The general balance condition

Let $P = \{x \in [l, u] : Ax \leq b, Ex = e\}$ be a nonempty compact polyhedron of contingent service tiers. The rows of A can include several discounted commitments at each history; the rows of E include information restrictions or an exact root payment commitment. Fix a feasible comparison protocol $z \in P$. Neither constancy nor interiority of z is assumed. All probabilities remain exogenous. Write

$$W_\lambda(x) = r(x) - \lambda T(x), \quad r(x) = p^\top x - \mathcal{R}_0(x), \quad T(x) = \sum_{a \in \mathcal{A}} \kappa_a |(Dx - h)_a|, \quad \lambda \geq 0. \quad (10)$$

Here $\mathcal{R}_0$ is convex and differentiable on a neighborhood of the box, $\kappa_a > 0$, and D is a specified incidence matrix. A temporal row is the difference between a tier and its parent's tier. An initial-installation row has one nonzero entry and its installed tier in h . Spatial switching or coordination edges may be included. Smooth graph coupling remains in $\mathcal{R}_0$. Omitted switching edges simply have no row. Let $I(z) = \{i : A_i z = b_i\}$, $g = \nabla r(z)$, and let $N_{[l,u]}(z)$ denote the outward box normal: its j th component is nonpositive at a lower boundary, nonnegative at an upper boundary, and zero in the interior. Fixed coordinates permit either sign.

Theorem 4.1 (Capacitated continuation balance). *The protocol z maximizes (10) on P if and only if there exist continuation prices $\mu_i \geq 0$ for $i \in I(z)$, unrestricted equality prices ν , a box normal $\xi \in N_{[l,u]}(z)$, and edge tensions s satisfying*

$$g = A_{I(z)}^\top \mu + E^\top \nu + \xi + D^\top s, \quad (11)$$

$$s_a = \lambda \kappa_a \operatorname{sign}(Dz - h)_a \quad \text{if } (Dz - h)_a \neq 0, \quad (12)$$

$$|s_a| \leq \lambda \kappa_a \quad \text{if } (Dz - h)_a = 0. \quad (13)$$

Consequently the set of friction coefficients for which this fixed protocol is optimal is a closed interval in $[0, \infty)$, possibly empty, a singleton, or unbounded. Its endpoints are the infimum and supremum of λ in the displayed linear program. These are linear programs in primitive derivatives, not the original nonlinear control problem.

Proof. The convex minimization form has objective $\mathcal{R}_0(x) - p^\top x + \lambda T(x)$. It is finite and continuous on a neighborhood of P . The subdifferential sum rule and the polyhedral normal formula give optimality exactly when $g \in \partial(\lambda T)(z) + N_P(z)$. The subgradients of each absolute value give (12)–(13); the polyhedral normal is $A_{I(z)}^\top \mu + E^\top \nu + \xi$. No strictly feasible point is needed for this normal representation. Conversely these equations and the supporting inequalities for $\mathcal{R}_0$, T , and P show $W_\lambda(x) \leq W_\lambda(z)$ for every feasible x . Finally, all conditions are linear in $(\lambda, s, \mu, \nu, \xi)$, including the signs of box normals. Their projection onto the scalar λ is a polyhedron, hence a closed interval with the stated possibilities. An unbounded endpoint is allowed. $\square$

The subgradient principle is standard convex analysis (Boyd and Vandenberghe, 2004); matrix absolute-value penalties and their dual treatment also appear in the generalized lasso (Tibshirani and Taylor, 2011). The economic content here is the separation of marginal operating reward into continuation prices, boundary scarcity, and bounded switching tensions. On a tree, the tensions can be eliminated entirely to obtain explicit subtree cuts.

4.2 Subtree cuts and an isotonic critical-friction program

For a scalar tree let $c_n > 0$ be discounted net-payment coefficients. Suppose the budget rows are $\sum_{j \geq n} c_j x_j \leq b_n$ for nonroot nodes and the root has exact payment equality. Let z be any feasible outside protocol. In this subsection D contains the temporal edges and initial installation: $h_o = q_0$, $h_n = 0$ off the root. Set

$z_{\text{pa}(o)} = q_0$. Introduce a node price ρ such that

$$\rho_o \in \mathbb{R}, \quad \rho_n - \rho_{\text{pa}(n)} \begin{cases} \geq 0, & n \in I(z), \\ = 0, & n \notin I(z), \end{cases} \quad (n \neq o). \quad (14)$$

Here $I(z)$ indexes binding continuation budgets only. Define the cumulative residual on a subtree by

$$H_n(\rho, \xi) = \sum_{j \succeq n} (g_j - c_j \rho_j - \xi_j). \quad (15)$$

Corollary 4.2 (Exact cut characterization). *The comparison protocol z is optimal if and only if some ρ satisfying (14) and some $\xi \in N_{[l,u]}(z)$ satisfy, at every node including the root,*

$$\begin{cases} H_n(\rho, \xi) = \lambda \kappa_n \text{sign}(z_n - z_{\text{pa}(n)}), & z_n \neq z_{\text{pa}(n)}, \\ |H_n(\rho, \xi)| \leq \lambda \kappa_n, & z_n = z_{\text{pa}(n)}. \end{cases} \quad (16)$$

For weak root acceptance replace the free root price by $\rho_o \geq 0$ when that budget binds, and by $\rho_o = 0$ when it is slack. If initial installation has no absolute cost, use $H_o = 0$ instead of the root capacity condition.

If $z_n = q_0 = \bar{\theta} \in (l_n, u_n)$ for every node, all continuation caps equal the payments of z , and all switching weights including the root are positive, the exact critical friction is finite and equals

$$\lambda_c = \min_{\rho: \rho_n \geq \rho_{\text{pa}(n)}} \max_n \frac{|\sum_{j \succeq n} (g_j - c_j \rho_j)|}{\kappa_n}. \quad (17)$$

The static protocol is optimal exactly for $\lambda \geq \lambda_c$. The root price in (17) is unrestricted under root equality.

Proof. The sum of the root multiplier and all binding-budget multipliers on a node's ancestral path is precisely ρ_n . This is equivalent to (14). For the rooted incidence matrix, $D^\top s = g - c\rho - \xi$ reads $s_n - \sum_{v: \text{pa}(v)=n} s_v = g_n - c_n \rho_n - \xi_n$. Summing over a subtree cancels all interior edges and leaves $s_n = H_n$. Substitution into Theorem 4.1 proves both directions, including installation and the weak-root variant. In the constant interior case there are no box normals or saturated edges. Minimizing the remaining capacities gives (17). Taking a constant ρ , for example zero, gives a finite feasible upper bound; the linear program attains its finite minimum. Increasing λ relaxes every remaining capacity, proving the ray characterization. $\square$

This program has a linear number of node variables and local balance constraints when subtree sums are represented recursively. Evaluating all cuts for given prices takes one backward tree pass. We do not assert that solving the linear program has linear running time. With $\lambda = 0$, the balance reduces to $g_n/c_n = \rho_n$, recovering the scalar edge criterion. For a two-review chain, eliminating its single payment-neutral transfer recovers the threshold in Corollary A.2. In contrast to that special case, (17) allows simultaneous transfers across all branches. For vector services, signed coefficients, several budgets, or extra graph edges, Theorem 4.1, rather than division by c_n , is the applicable formulation.

4.3 Constructive improvements and comparative statics

Let $K = T_P(z)$ be the polyhedral tangent cone. Set $v = Dz - h$ and

$$\phi_z(d) = \sum_{a:v_a \neq 0} \kappa_a \operatorname{sign}(v_a)(Dd)_a + \sum_{a:v_a=0} \kappa_a |(Dd)_a|.$$

Thus ϕ_z is the directional derivative of switching cost, including boundary installation. The following alternative gives an implementable improving policy when the capacity test fails.

Proposition 4.3 (Certified direction and friction sensitivity). *The balance test fails exactly when the piecewise-linear program*

$$\delta = \max_{d \in K, \|d\|_\infty \leq 1} \{g^\top d - \lambda \phi_z(d)\} \quad (18)$$

has positive value. Absolute-value epigraphs turn it into a linear program. For a maximizing direction d , choose $a_{\max} > 0$ so that $z + ad \in P$ and no nonzero component of $Dz - h$ changes sign for $0 \leq a \leq a_{\max}$. If the smooth resource curvature along this segment is at most L_d , then

$$W_\lambda(z + ad) - W_\lambda(z) \geq a\delta - L_d a^2/2. \quad (19)$$

When $T(z) = 0$, the accepted gain $G(\lambda) = \max_{x \in P} W_\lambda(x) - r(z)$ is nonnegative, nonincreasing, and convex in λ . If x_λ is an optimizer, $-T(x_\lambda)$ is a subgradient of G ; optimal total switching is nonincreasing with friction. Tightening any commitments while keeping z feasible cannot increase either G or its critical friction. Increasing edge weights cannot increase the critical friction of a zero-switching comparator.

Proof. For a convex objective on a polyhedron, nonnegative directional derivatives in every tangent direction are equivalent to global minimality. Negating this statement gives (18); homogeneity permits the bounded normalization. A tangent direction is feasible for a sufficiently short segment. Bounds from inactive inequalities, box faces, and the finitely many nonzero v_a give the required positive $a_{\max}$. On that segment $T(z + ad) - T(z) = a\phi_z(d)$ exactly. Integrating the smooth curvature bound proves (19); take $a = \min\{a_{\max}, \delta/L_d\}$ if $L_d > 0$. For $T(z) = 0$, G is a supremum of affine nonincreasing functions of λ , and z supplies the zero lower bound. The maximizing affine function supplies its subgradient. Adding the two optimality inequalities at two friction values orders their optimal switching costs. Feasible-set inclusion proves the commitment statement. Larger edge weights reduce every nonstatic reward without changing $r(z)$, proving the final statement. $\square$

For a zero-switching comparator the equivalent variational threshold is the positive part of $\sup_{d \in K, T'(z;d) > 0} g^\top d / T'(z;d)$. A direction with $T'(z;d) = 0$ and $g^\top d > 0$ makes this threshold infinite. The initial edge with positive weight eliminates such directions on a scalar rooted tree. This convention matters when installation or an entire component is unpenalized.

For a nonconstant comparison protocol, it would be incorrect to extrapolate the monotone-friction conclusion. The relative gain is still convex, but its supporting slope is $T(z) - T(x_\lambda)$, which can have either sign. An exact example fixes a root tier at zero and permits a child tier $x \in [0, 1]$, with $r(x) = x - (x - 1)^2/2$

and $T(x) = x$. The outside protocol $z = 1$ is optimal exactly on $0 \leq \lambda \leq 1$. For $1 < \lambda < 2$ the optimizer is $2 - \lambda$, and for $\lambda \geq 2$ it is zero. Increasing friction therefore makes this already-changing outside protocol improvable. The interval assertion of Theorem 4.1, rather than a universal lower threshold, covers this case.

4.4 A sharp price–capacity margin for the value of adaptation

The balance test also yields a quantitative measure of accepted value. This measure optimizes the certificate’s prices, rather than choosing arbitrary multipliers and interpreting a possibly loose bound. For a positive definite metric H , define

$$C_\lambda(z) = N_P(z) + \partial(\lambda T)(z), \quad \Delta_\lambda = \min_{c \in C_\lambda(z)} \|g - c\|_{H^{-1}}. \quad (20)$$

This is a convex quadratic program in the same continuation prices and edge tensions as the linear balance test. It measures the marginal reward that cannot be absorbed by commitments, boundaries, and switching capacities.

Theorem 4.4 (Sharp price–capacity margin). *Suppose $\mathcal{R}_0$ is m_0 -strongly convex and L_0 -smooth in metric H , with $0 < m_0 \leq L_0$. Let c^* attain (20) and put $d = H^{-1}(g - c^*)$. Then $d \in T_P(z)$ and*

$$g^\top d - \lambda \phi_z(d) = \|d\|_H^2 = \Delta_\lambda^2. \quad (21)$$

For any $a_{\max} > 0$ such that the segment $z + [0, a_{\max}]d$ is feasible and preserves every nonzero switching sign, write $a = \min\{a_{\max}, 1/L_0\}$. The exact accepted gain satisfies

$$(a - L_0 a^2/2) \Delta_\lambda^2 \leq \max_{x \in P} W_\lambda(x) - W_\lambda(z) \leq \frac{\Delta_\lambda^2}{2m_0}. \quad (22)$$

If $a_{\max} \geq 1/L_0$, the lower bound is $\Delta_\lambda^2/(2L_0)$. If the smooth resource is a quadratic with Hessian $m_0 H$ and $a_{\max} \geq 1/m_0$, both bounds coincide and $z + d/m_0$ is globally optimal. Thus the same price–capacity program gives both the threshold and the exact gain on this quadratic subclass.

Proof. The set $C_\lambda(z)$ is a nonempty closed polyhedron, so the metric projection exists. Write $c^* = n^* + t^*$ with $n^* \in N_P(z)$ and $t^* \in \partial(\lambda T)(z)$. The projection variational inequality gives $d^\top(c - c^*) \leq 0$ for all $c \in C_\lambda(z)$. Varying only the normal component and taking it equal to zero or $2n^*$ gives $d^\top n^* = 0$; varying it arbitrarily gives $d \in N_P(z)^\circ = T_P(z)$. Varying only the switching component gives $d^\top t^* = \sup_{t \in \partial(\lambda T)(z)} d^\top t = \lambda \phi_z(d)$. Since $(g - c^*)^\top d = \|d\|_H^2$, this proves (21).

For any feasible x , the normal inequality, the switching supporting inequality, and strong convexity bound its gain by $(g - c^*)^\top(x - z) - m_0\|x - z\|_H^2/2$. Completing the square gives the upper bound. On the specified segment switching cost has its directional affine form. Smoothness and (21) give the lower bound at $z + ad$. When the smooth resource is the stated quadratic and $a = 1/m_0$, the lower bound reaches the global upper bound. This also proves the claimed policy’s global optimality. $\square$

For zero-switching comparators, the critical friction is the smallest λ for which $\Delta_\lambda = 0$. A small

nonzero margin with limited box clearance can correspond to a much smaller realizable gain; (22) keeps that clearance explicit. With a nonconstant comparator, sign clearance also matters. These qualifications prevent interpreting an unconstrained quadratic distance as the accepted gain when a proposed step crosses a tier boundary or changes a pre-existing switching sign.

Budget values and normalization. For fixed dynamics, objective, and equality constraints, the optimal value is concave and nondecreasing as a function of feasible upper-budget vectors b . Mixing two feasible policies proves concavity; an optimal nonnegative budget multiplier is a supergradient by weak duality. These are sensitivity statements for externally fixed tariffs, not endogenous tariff-design results. In the recursion (53), b and $\bar{b}_n$ are *conditional* remaining discounted payments, measured from node n . Multiplication by $w_n = \beta^{t(n)} \Pr(n)$ gives the unconditional subtree budget used in A . The child term uses βP_{nv} , not the unconditional child probability a second time. Several commitments add a vector budget state and one such inequality per commitment. Resource and promise states are classical constructions (Altman, 1999; Spear and Srivastava, 1987; Thomas and Worrall, 1988); the present contribution is the continuation-price and switching-capacity structure they induce in this observable service-control problem.

4.5 Accepted transfers as coordinates, not a repair heuristic

The scalar specialization has a useful algorithmic consequence. In this subsection the outside tier is constant and interior, $z_n = q_0 \in (l_n, u_n)$, payment coefficients satisfy $c_n > 0$, every nonroot subtree has its comparator payment cap, and the root payment is fixed at the comparator level. Other polyhedra remain covered by Theorem 4.1; the coordinate construction below uses these additional assumptions. Include the root installation edge in D , so that $D(x - z)$ contains every change from the installed protocol.

For a nonroot node n , define the unpaid continuation amount and its tier-scaled version by

$$f_n = - \sum_{j \succeq n} c_j (x_j - z_j), \quad y_n = f_n / c_n.$$

Let B have one column for each nonroot node: $B_{nn} = -1$, $B_{\text{pa}(n),n} = c_n / c_{\text{pa}(n)}$, and all other entries zero. Column indices here name nodes rather than consecutive integers.

Proposition 4.5 (Exact continuation-flow coordinates). *The map $x = z + By$ is a bijection between the scalar accepted polytope and*

$$\mathcal{Y} = \{y \geq 0 : l - z \leq By \leq u - z\}. \quad (23)$$

It preserves all nonroot continuation caps and the root payment equality. Both the map and its inverse require $O(N)$ arithmetic operations and storage on an N -node tree. In particular, optimizing in these coordinates does not restrict the feasible set to a selected collection of transfers.

Proof. Set $f_o = 0$ at the root. Subtracting child subtree sums from the subtree sum at n gives $c_n(x_n - z_n) = \sum_{j:\text{pa}(j)=n} f_j - f_n$. This is exactly $x - z = By$. Feasibility gives $f_n \geq 0$. Conversely, summing this identity over any subtree cancels every internal transfer, leaving $-f_n$; summing over the whole tree gives zero.

Bounds on x give precisely the two remaining inequalities in (23). The same identities prove uniqueness. One backward summation computes all f_n ; B has two entries per column. $\square$

An amount f_n moves payment from the continuation rooted at n to its parent without changing total payment. Its nonnegativity records the customer's exit option, not an artificial restriction on a learned actor. Conditional and unconditional subtree constraints are equivalent because the conditioning probability is positive.

4.6 A constructive critical-friction bracket

Put $g = \nabla r(z)$, $a = B^\top g$, and $K = DB$. For $y \neq 0$ define $T_B(y) = \sum_n \kappa_n |(Ky)_n|$. Since D is invertible and B has full column rank, $T_B(y) > 0$. The tangent cone of (23) at zero is the nonnegative orthant. Thus the critical friction admits the primal and dual expressions

$$\lambda_c = \max \left\{ 0, \sup_{y \geq 0, y \neq 0} \frac{a^\top y}{T_B(y)} \right\} = \min \{ \lambda \geq 0 : K^\top s \geq a, |s_n| \leq \lambda \kappa_n \}. \quad (24)$$

The first equality follows from the directional optimality condition; the second follows from the support function of the tension box and separation. Equivalently, minimize λ subject to

$$v = D^\top s, \quad B^\top v \geq a, \quad -\lambda \kappa \leq s \leq \lambda \kappa.$$

Keeping v avoids forming DB . This linear program has $2N + 1$ variables and $9N - 3$ matrix nonzeros with the displayed representation. It is not a linear-time claim for solving a general linear program.

Theorem 4.6 (Two-pass repair and an improving decision). *Take any $\lambda \geq 0$ and any s with $|s_n| \leq \lambda \kappa_n$. Let $r = (a - B^\top D^\top s)_+$. Define $v_o = 0$, and in forward tree order set*

$$v_n = \frac{c_n}{c_{\text{pa}(n)}} v_{\text{pa}(n)} - r_n, \quad d_n = \sum_{j \geq n} v_j.$$

Then, for every nonzero $y \geq 0$,

$$\max \{ 0, a^\top y / T_B(y) \} \leq \lambda_c \leq \max_n \frac{|s_n + d_n|}{\kappa_n} \leq \lambda + \max_n \frac{|d_n|}{\kappa_n}. \quad (25)$$

All quantities and both feasibility witnesses are computable in $O(N)$ arithmetic operations. If $\Gamma = a^\top y - \lambda T_B(y) > 0$ and the smooth resource is quadratic with Hessian Q , let $q = y^\top B^\top Q B y > 0$ and $t_b = \sup \{ t \geq 0 : z + t B y \in [l, u] \}$. The accepted policy with $t = \min \{ t_b, \Gamma / q \}$ earns exactly

$$W_\lambda(z + t B y) - W_\lambda(z) = t \Gamma - t^2 q / 2 > 0. \quad (26)$$

Proof. The forward recursion gives $B^\top v = r$. Backward subtree summation gives $D^\top d = v$, including the installation edge. Consequently $B^\top D^\top (s + d) = B^\top D^\top s + r \geq a$. This proves the upper bound by dual

feasibility; the last inequality is the triangle inequality. The lower bound is the first expression in (24). Every operation is a tree traversal. Interior bounds give $t_b > 0$. Along $z + tBy$, switching is positively homogeneous because all changes vanish at z , whereas the smooth quadratic expands exactly. Maximizing the resulting concave quadratic on $[0, t_b]$ proves (26). No additional switching-sign clearance is needed along this ray. $\square$

The repaired tension has an immediate price interpretation. For $\alpha = B^\top D^\top (s + d) - a \geq 0$, the continuation-price increment at node n is α_n / c_n . The root price is unrestricted. These increments reconstruct the supporting ancestor prices of Corollary 4.2. A positive lower witness instead identifies the branches on which accepted payment reallocation can finance profitable reconfiguration.

A sparse first-order implementation. Write $s = \text{diag}(\kappa)t$ and $b = \text{diag}(c_{-o})^{-1}a$, and define

$$M = \text{diag}(c_{-o})^{-1} B^\top D^\top \text{diag}(\kappa).$$

At a trial friction minimize

$$\phi_\lambda(t) = \frac{1}{2} \|(b - Mt)_+\|_2^2, \quad |t_n| \leq \lambda. \quad (27)$$

Its gradient is $-M^\top (b - Mt)_+$ and is Lipschitz with constant at most $L = \|M\|_1 \|M\|_\infty$. Accelerated projected gradient therefore has the standard $O(L\|t_0 - t_*\|^2/k^2)$ objective bound (Beck and Teboulle, 2009). Products with M and its transpose use the two sparse factors, not a materialized matrix. Their absolute row and column sums also have linear-time formulas, given in the companion. Each iteration is $O(N)$; neither the conditioning nor the required number of iterations is asserted to be dimension independent.

Apply Theorem 4.6 to intermediate tensions, using $y = \text{diag}(c_{-o})^{-1}(b - Mt)_+$ as a lower witness when nonzero. Maintain the best lower and upper bounds and choose the next trial at their midpoint. If a trial is infeasible, its minimizer has residual $r \neq 0$. The box optimality inequality gives $r^\top Mt = \lambda \|M^\top r\|_1$ and hence $b^\top r = \lambda \|M^\top r\|_1 + \|r\|^2$. Thus its lower witness strictly exceeds the trial friction. For a feasible trial the residual tends to zero and the repair upper bound tends to a value no greater than that trial. These facts explain the bracket algorithm without interpreting a small gradient norm as an exact feasibility decision. The implementation stops on the bracket width or reports an iteration cap; it stores rational witnesses in either case.

The computational contribution is the continuation-preserving coordinate map, explicit tension repair, and accepted-gain witness. Total-variation duality and regularization paths themselves are established (Tibshirani and Taylor, 2011; Arnold and Tibshirani, 2016). Unlike unconstrained fused-lasso denoising, the admissible directions here form a payment-transfer cone; its polar inequality is one-sided and determines continuation prices. Sparse generic linear programming remains an important, and often faster, comparator.

5 What accepted adaptivity buys

The occupation program establishes a feasible improvement, but an information decomposition must impose its commitments in every class. We now solve that comparison with continuous tiers, rather than transferring

an unconstrained information increment to the accepted problem. A supporting-price argument first identifies the physical policy from the commitments themselves.

5.1 The service and resource commitments identify stock

Lemma 5.1 (Physical-policy identification). *Let s_z be a fixed-stock rule and suppose that, for some $\xi > 0$,*

$$C_z(S) - C_z(s_z) - \xi\{f_z(S) - f_z(s_z)\} \geq 0 \quad (28)$$

for every feasible z, S , with equality only at $S = s_z$. Demand regimes are exogenous. Any protocol with $F^\pi \geq F^s$ and $C^\pi \leq C^s$ must select $S_t = s_{z_t}$ almost surely at every positive-probability review. Both commitments then hold with equality. For the canonical instance, $s_z = z + 2$ and $\xi = 1$ satisfy (28), for both the original stock menu and every stock in $[0, 6]$.

Proof. The expected discounted sum of the left side of (28) is nonnegative. Feasibility bounds that sum above by $(C^\pi - C^s) - \xi(F^\pi - F^s) \leq 0$. Every nonnegative summand at a reachable state must therefore vanish. Strictness identifies stock, and exogeneity gives equality of both totals. In the instance, $C_z(z+2) = 1.03+0.3z$ and $f_z(z+2) = 1.5+z$. On the half-unit menu, the smallest positive supporting gap is exactly $9/40$. The functions are piecewise linear between the demand atoms, all of which belong to that menu. The same unique minimizer consequently holds on the entire stock interval. The positive gap $9/40$ is a menu gap, not a uniform gap on continuous stocks approaching $z+2$. $\square$

This is not unconstrained stock elimination inside a constrained problem. Both the service floor and the physical-cost cap are used to prove that no other stock decision is feasible. Under looser commitments, we retain all stock choices in the occupation program.

Set $w_{tz} = \beta^t \Pr(z_t = z)$, $a_z = r_z - \kappa L_z(s_z)$, and $A = \sum_{t,z} w_{tz} a_z$. At the canonical stocks, $a_z = 0.36+1.4z$. Let $\bar{\theta}$ be the optimized continuous static tier, and put $\bar{P} = A\bar{\theta}$. The remaining acceptance condition is a payment budget,

$$P^\pi := \mathbb{E}^\pi \sum_t \beta^t a_{z_t} \theta_t \leq \bar{P}, \quad U^\pi = \nu F^s - P^\pi. \quad (29)$$

The accepted reward is $W^\pi = P^\pi - C^s - \mathbb{E}^\pi \sum_t \beta^t [m\theta_t^2 + \lambda(\theta_t - q_t)^2]$. Thus matching the customer's utility fixes expected net payments. The remaining economic benefit must be a reduction in maintenance and reconfiguration resources, not an unreported transfer from the customer.

5.2 Randomization and the accepted information hierarchy

We define randomization before comparing information. An ex ante public random tape R , independent of demand regimes, may select a constant tier, a time schedule, or a map of time and current regime. These classes have rules $f(R)$, $f_t(R)$, and $f_t(z, R)$, respectively. They include deterministic rules and public mixtures of them. They do not silently grant a regime-only rule access to earlier regimes through an unrecorded memory state. Full protocols may use the entire observed history; their optimum will admit a Markov implementation.

Proposition 5.2 (Derandomization on a continuous accepted menu). *Under Lemma 5.1, continuous tier feasibility $[0, 1]$, and convex maintenance and adjustment costs, public randomization does not improve the accepted static, time-only, or time–regime value. An inherited-only randomized rule with deterministic initial tier has the same accepted value as a time-only rule. For the canonical quadratic problem, a deterministic full-state rule also attains the unrestricted accepted optimum.*

Proof. Replace each restricted rule by its expectation over R , holding the regime path fixed. The resulting action remains in $[0, 1]$ and in the same information class. Expected payments do not change. Jensen’s inequality decreases each maintenance cost and each adjustment cost, including the jointly randomized difference of consecutive tiers. The fixed physical policy preserves the other commitments. Without regime inputs, an inherited-only rule generates a random time schedule from q_0 and R ; the preceding argument applies. For full histories the conditional-mean construction need not preserve the displayed Markov parametrization. The deterministic Markov rule and a matching unrestricted Lagrangian upper bound constructed below establish attainment directly. None of these statements asserts derandomization on a coarse, nonconvex tier menu. $\square$

For a multiplier $\eta \geq 0$, maximize $W^\pi - \eta P^\pi$ and add $\eta \bar{P}$. This is a different sign convention from the customer-utility multiplier in (EC.5), with the constant customer-service term removed. Write its full-state value as

$$J_t^\eta(z, q) = -A_t q^2 + B_{tz} q + D_{tz}, \quad A_T = B_{Tz} = D_{Tz} = 0. \quad (30)$$

When the maximizers are interior, backward completion of squares gives

$$\begin{aligned} K_t &= m + \lambda + \beta A_{t+1}, & L_{tz} &= (1 - \eta) a_z + \beta \sum_{z'} P_{zz'} B_{t+1, z'}, \\ A_t &= \lambda - \lambda^2 / K_t, & B_{tz} &= \lambda L_{tz} / K_t, \\ D_{tz} &= -C_z(s_z) + \beta \sum_{z'} P_{zz'} D_{t+1, z'} + L_{tz}^2 / (4K_t), & \theta_t^\eta(z, q) &= \lambda q / K_t + L_{tz} / (2K_t). \end{aligned} \quad (31)$$

Endpoint verification is essential: the affine rule must lie in $[0, 1]$ for every $q \in [0, 1]$, not just along one simulated path. We verify both endpoints in rational arithmetic for every period and regime in the canonical certificate.

The restricted classes require their own optimization. Give each reachable time–information pair one tier variable x_i . Their rewards have the form

$$W(x) = -x^\top H x + b^\top x - c_0, \quad c^\top x \leq \bar{P}, \quad H \succ 0. \quad (32)$$

Here the maintenance terms contribute $m w_i$ to the diagonal; an adjustment edge from variable j to variable i contributes its discounted transition weight times $\lambda(x_i - x_j)^2$. The initial installation cost contributes $\lambda(x_0 - q_0)^2$. There are one, eight, and twenty-two variables in the static, time, and time–regime problems.

For an interior solution,

$$x = \frac{1}{2}H^{-1}(b - \eta c), \quad \eta = \max \left\{ 0, \frac{c^\top H^{-1}b - 2\bar{P}}{c^\top H^{-1}c} \right\}.$$

The matrix is not a full-state Bellman relaxation of the regime restriction: a single variable is shared by every history with the same permitted information.

Theorem 5.3 (Exact accepted information values). *For the canonical primitives, the commitments of the optimized continuous static contract, and continuous stock and tier intervals, the accepted values in Table 1 are exact rational numbers. Each value is attained without public randomization. The accepted increment from observing the inherited tier, after time and current regime have already been admitted, is 0.019290378765 The total gain over the optimized static contract is 0.636665271611*

Proof. The exhaustive static stock-envelope calculation yields the stated $\bar{\theta}$ and stocks. Lemma 5.1 then reduces every feasible protocol, not just the proposed ones, to (29). For each restricted class, the displayed solution of (32) has every tier strictly between zero and one. Rational arithmetic verifies its stationarity equation, nonnegative multiplier, primal feasibility, and complementary slackness. Strict concavity proves global optimality in the deterministic class; Proposition 5.2 includes its public mixtures.

In the full problem, A_t is independent of η and B_{tz} is proportional to $1 - \eta$. Forward first moments make P^{θ^η} affine in η . Solving $P^{\theta^\eta} = \bar{P}$ gives $\eta = 0.180271243733$ Rational endpoint checks verify interior maximization in every Bellman step. The affine policy is feasible, and its Lagrangian bound is equal to its reward. Forward second moments evaluate maintenance and adjustment exactly. This bounds every randomized history-dependent protocol from above and attains the bound with a deterministic Markov policy. The files record the rational values, multipliers, and policy coefficients, rather than only the decimals in the table. $\square$

The inherited-tier increment is approximately three percent of the total accepted gain. Most of the gain is due to current-regime reallocation; attributing the entire improvement to contractual memory would be incorrect. Nevertheless, inherited information has strictly positive value under the same customer commitments, not only in a provider relaxation. The continuous optimum also exceeds the certified 0.1-menu optimum by 0.011524959639 The earlier lottery is therefore not the source of the economic improvement.

5.3 A primitive mechanism for a robust improvement

Theorem 5.4 (A customer-neutral adaptive direction). *Consider a feasible constant tier $\bar{\theta} \in (0, 1)$ with fixed stocks s_z , linear net payments $a_z\theta$, and twice continuously differentiable maintenance M near $\bar{\theta}$, with $M'(\bar{\theta}) > 0$. Suppose that at a review $t \geq 1$ two regimes have positive discounted weights w_L, w_H and $0 < a_L < a_H$. For every finite quadratic adjustment coefficient, a sufficiently small deterministic time-regime perturbation strictly improves provider reward while preserving customer utility, fill, and physical cost exactly. The conclusion holds in a neighborhood of primitives preserving these strict inequalities.*

Proof. Change only that review's tiers by εd_z , where

$$d_H = (w_H a_H)^{-1}, \quad d_L = -(w_L a_L)^{-1}, \quad d_z = 0 \text{ otherwise.}$$

Then $\sum_z w_z a_z d_z = 0$ and $\sum_z w_z d_z = 1/a_H - 1/a_L < 0$. Payments and physical decisions are unchanged in expectation. The first-order reward gain is $\varepsilon M'(\bar{\theta})(1/a_L - 1/a_H) > 0$. All changed adjustment costs are second order because the baseline tier is constant at this and the next review; the initial installation is unaffected. If $|M''| \leq L_M$ locally, the absolute quadratic cost is at most

$$\varepsilon^2 \{L_M/2 + \lambda + \beta \lambda \mathbf{1}_{\{t < T-1\}}\} \left(\frac{1}{w_H a_H^2} + \frac{1}{w_L a_L^2} \right).$$

Choose ε small enough to keep both tiers interior and make this term smaller than the positive linear gain. Continuity preserves a strictly positive margin in a sufficiently small neighborhood. For $M(\theta) = m\theta^2$, the displayed first- and second-order terms give the exact payoff difference, not just an asymptotic expansion. $\square$

This argument does not require translated demand, a coarse menu, or a lottery. It does require positive and heterogeneous net payment rates and an interior static tier with increasing maintenance cost. It proves existence of an improvement, not that every parameter perturbation has a large gain. It also explains the role of the customer constraint: concentrating a fixed expected payment budget on higher-payment regimes permits a lower average maintained tier. Reconfiguration is worthwhile when those savings exceed its resource cost.

5.4 Allowing the customer to leave at each review

The ex ante institution can be weakened without introducing hidden action or changing the tariff. At every observed regime history h at review t , let the customer's outside option be the remaining service utility of the same static contract. The provider remains committed to the announced protocol, but the customer may leave at each such review. Physical stock is still identified by the service and cost commitments. Continuation participation therefore requires

$$\mathbb{E} \left[\sum_{s=t}^{T-1} \beta^{s-t} a_{z_s} \theta_s \middle| h \right] \leq \bar{\theta} \mathbb{E} \left[\sum_{s=t}^{T-1} \beta^{s-t} a_{z_s} \middle| h \right] \quad \text{for every reachable } h. \quad (33)$$

These constraints cannot be replaced by the root participation inequality. On the finite regime tree they are linear, while maintenance and adjustment give a strictly concave quadratic objective. The canonical tree has 3,280 decision nodes. Its policy may depend on history; a promise-state representation is an alternative, not an assumption that the original three-coordinate Markov policy remains sufficient.

We solve this convex program numerically, round its deterministic tiers to rational numbers, and contract the vector slightly toward zero to enforce every continuation inequality exactly. The resulting reward is $-5.417032209823 \dots$, an improvement of $0.453078763224 \dots$ over the static contract. A rational feasible-policy and weighted-residual certificate places the interim-participation optimum less than 6.95×10^{-7} above

this reward. Thus the exit option costs approximately 0.18358651 relative to full customer commitment, but does not eliminate the gain. EC Section EC.3 gives the upper-bound proof and the complete policy audit. No compensating side payment or change in the fixed menu is used to obtain this result.

6 Learning on acceptance faces and certified deployment

The structural balance condition also provides a common language for learned and classical implementations. This section distinguishes three questions: whether a candidate is feasible, how far its reward can be from optimal, and whether learning selects candidates with useful out-of-sample guarantees. None of these questions is answered by a solver's success flag.

6.1 The same balance equations certify a complete contingent policy

Suppose $\mathcal{R}_0$ is m_0 -strongly convex in a positive definite metric H , that is, its supporting quadratic has term $m_0\|y - x\|_H^2/2$. A numerical candidate x must first belong to P . Choose any $\mu \geq 0$, any equality price ν , nonnegative upper and lower bound prices u^+, u^- , and edge tensions s satisfying (12)–(13) at x , so that $D^\top s \in \partial(\lambda T)(x)$. Define

$$\begin{aligned} e_x &= \nabla r(x) - D^\top s - A^\top \mu - E^\top \nu - u^+ + u^-, \\ C(x) &= \frac{\|e_x\|_{H^{-1}}^2}{2m_0} + \mu^\top (b - Ax) + (u^+)^\top (u - x) + (u^-)^\top (x - l). \end{aligned} \quad (34)$$

The vector u without a superscript denotes the upper tier bound, not a multiplier.

Theorem 6.1 (Balance-residual policy certificate). *Every quantity in (34) is nonnegative except possibly individual residual coordinates, and*

$$0 \leq \max_{y \in P} W_\lambda(y) - W_\lambda(x) \leq C(x).$$

For polynomial smooth resources and rational primitives, candidate, prices, and tensions, feasibility and this bound are exactly checkable when H^{-1} and m_0 are rational. This certificate covers boundary policies and absolute switching. Its zero-residual, zero-complementarity condition is exactly Theorem 4.1 at the candidate.

Proof. Strong concavity of r and the supporting inequality for λT bound $W_\lambda(y) - W_\lambda(x)$ above by $(\nabla r(x) - D^\top s)^\top (y - x) - m_0\|y - x\|_H^2/2$. Substitute the residual. Feasibility bounds the budget contribution by $\mu^\top (b - Ax)$ and the box contributions by the two displayed slacks. The equality contribution vanishes. Completing the square bounds the remaining residual term by $\|e_x\|_{H^{-1}}^2/(2m_0)$. All inputs of the resulting polynomial/rational expression can be replayed without trusting the candidate generator. $\square$

This is a specialization of the standard strong-convexity bound, not a claim that residual inequalities were invented here. Unlike a stand-alone unconstrained gradient norm, it retains the continuation and boundary

prices that also determine accepted adaptivity. In the quartic experiment $H = \text{diag}(w)$ and $m_0 = 2m$, recovering the earlier certificate and adding explicit box prices.

For a feasible comparison protocol z , let $\hat{x}$ be whichever of x, z has larger exactly evaluated reward. A valid gap for the selected decision is

$$\min\{W_\lambda(x) + C(x), W_\lambda(z) + C(z)\} - W_\lambda(\hat{x}). \quad (35)$$

The use of both upper bounds is optional but can be substantially tighter than carrying the failed proposal's bound through the gate. A prescribed tolerance is met only when the selected decision's bound meets it. Otherwise the deployed algorithm must refine, return a separately certified alternative, or explicitly report that the tolerance was not attained.

6.2 Why the acceptance face matters for gradient learning

Consider a fixed feasible set and define the forcing-value function $J(p) = \max_{x \in P} \{p^\top x - \mathcal{R}_0(x) - \lambda T(x)\}$. Strong convexity gives a unique optimizer $x^*(p)$ and $\nabla J(p) = x^*(p)$. For the weighted forcing used in the experiments, $\nabla_p J = \text{diag}(w)x^*$ instead. A scalar potential therefore produces compatible value and policy estimates, but compatibility alone does not make the estimated actor feasible.

Let μ^*, ν^*, ξ^*, s^* be optimal prices and tensions at x^* . Define the convex exposed set

$$\begin{aligned} \mathcal{F}_* = \{y \in P : \mu^{*\top}(b - Ay) = 0, \quad \xi^{*\top}(x^* - y) = 0, \\ \lambda T(y) - \lambda T(x^*) - s^{*\top}D(y - x^*) = 0\}. \end{aligned} \quad (36)$$

The last equality selects the face of the piecewise-linear switching cost supported by s^* . It is automatically satisfied when $\lambda = 0$.

Theorem 6.2 (Gradient accuracy, acceptance prices, and policy loss). *For every feasible y ,*

$$\begin{aligned} J(p) - [p^\top y - \mathcal{R}_0(y) - \lambda T(y)] &= D_{\mathcal{R}_0}(y, x^*) + \mu^{*\top}(b - Ay) + \xi^{*\top}(x^* - y) \\ &\quad + \lambda T(y) - \lambda T(x^*) - s^{*\top}D(y - x^*). \end{aligned} \quad (37)$$

All four terms are nonnegative. Suppose additionally that $\mathcal{R}_0$ has L_0 -Lipschitz gradient in metric H on P . For any predicted actor a , its metric projection $y = \Pi_{\mathcal{F}_}^H(a)$ satisfies*

$$0 \leq J(p) - W_\lambda(y) \leq \frac{L_0}{2} \|a - x^*\|_H^2. \quad (38)$$

In particular, for a scalar compatible critic use $a = \nabla \hat{J}(p)$ (or the appropriately weight-normalized gradient). For any distribution on problems with uniformly bounded L_0 , the same inequality holds after expectation whenever these face conditions hold problem by problem.

Proof. Add and subtract the smooth linearization at x^* and use $p - \nabla \mathcal{R}_0(x^*) = A^\top \mu^* + E^\top \nu^* + \xi^* + D^\top s^*$. Complementary slackness and $E(y - x^*) = 0$ give (37). Convexity, feasibility, the outward normal, and

the switching subgradient give the four signs. On $\mathcal{F}_*$ only the smooth Bregman term remains, bounded by $L_0\|y - x^*\|_H^2/2$. Projection onto a closed convex set containing x^* is nonexpansive relative to x^* , so $\|y - x^*\|_H \leq \|a - x^*\|_H$. Taking expectations proves the final assertion. $\square$

This theorem is an approximation-to-decision result, not an assertion that the unknown optimal face is available for free. Verified optimal prices identify it; an unverified predicted face does not establish (38). The residual certificate remains valid without knowing it. The decomposition explains why a common scaling repair can lose reward even when actor distance is small: it can introduce first-order slack in expensive continuation commitments. Indeed, on $[0, 1]$, maximizing $x - (x - 1)^2/2$ gives $x^* = 1$, while the feasible actor $1 - \epsilon$ loses $\epsilon + \epsilon^2/2$, not merely a quadratic amount. Ignoring the boundary-price term would yield a false gradient-to-regret theorem. The full-face projection in this example is the singleton $\{1\}$.

6.3 A finite-sample certificate for selecting a learning pipeline

The preceding results do not certify arbitrary distribution shift. They can, however, turn independent validation into a precise learning statement. Suppose M frozen candidate-generation pipelines, including all repair, fallback, and refinement rules, are fixed before a validation sample is drawn. The pipelines may have been fitted on separate training data. On problem I , pipeline j returns a feasible policy with regret bounded by a measurable certificate $C_j(I) \in [0, B]$. A uniform range bound B must be established from the problem class, not obtained by discarding large observed bounds. When a separate valid range bound on regret is available, taking its minimum with a residual certificate is legitimate.

Proposition 6.3 (Certificate-risk selection). *For n independent validation problems drawn from the same distribution as a future problem, with probability at least $1 - \delta$ simultaneously for all M pipelines,*

$$\mathbb{E} \text{Regret}_j(I) \leq \frac{1}{n} \sum_{i=1}^n C_j(I_i) + B \sqrt{\frac{\log(M/\delta)}{2n}}. \quad (39)$$

The statement also holds for the pipeline selected by minimizing the empirical certificate mean. Moreover its expected certificate is at most the best pipeline's expected certificate plus $2B\sqrt{\log(2M/\delta)/(2n)}$ with probability at least $1 - \delta$.

Proof. Condition on the training data and hence on the frozen pipelines. Apply the one-sided bounded-variable Hoeffding inequality to each certificate mean and take a union bound. Expected regret is no larger than expected certificate. Simultaneity permits validation-based selection. For the final assertion use the two-sided inequality, then compare the selected empirical mean to that of a population-minimizing pipeline and add the two deviations. Unconditioning preserves both probability statements. $\square$

The concentration argument is standard (Hoeffding, 1963); the certified object is the selected *complete decision pipeline*, not a neural architecture or an unrestricted distribution-shift claim. A validation set used to change features or tolerances no longer qualifies as independent validation for the displayed guarantee. The predecessor computational study retained in the companion is a crossed stress test, not an independent-validation experiment. Section 7 executes independent validation for the frozen verified-cell policies.

Constrained-policy learning and value-based safety constructions study different sources of uncertainty (Satija et al., 2020; Wachi and Sui, 2020); here the feasible polyhedron is known and deterministic certification is separated from candidate quality.

6.4 Approximate prices without an unknown optimal face

The face in Theorem 6.2 explains a loss mechanism but does not by itself implement a policy. We now remove that oracle from the certificate and then give a verifiable local policy representation. First permit *arbitrary* predicted tensions satisfying only $|s_a| \leq \lambda \kappa_a$. They need not have the candidate's switching signs. Define their switching leakage by

$$\Psi_s(x) = \lambda T(x) - s^\top (Dx - h) \geq 0.$$

Keep the residual e_x in (34), with arbitrary nonnegative inequality and box prices and an arbitrary equality price. Let

$$\mathcal{L}_x = \mu^\top (b - Ax) + (u^+)^\top (u - x) + (u^-)^\top (x - l) + \Psi_s(x). \quad (40)$$

Theorem 6.4 (Computable approximate-face certificate). *For every feasible x and every such set of predicted prices,*

$$0 \leq W_\lambda(x^*) - W_\lambda(x) \leq \tilde{C}(x) := \frac{\|e_x\|_{H^{-1}}^2}{2m_0} + \mathcal{L}_x. \quad (41)$$

In particular, residual norm at most ϵ and leakage at most η give regret at most $\epsilon^2/(2m_0) + \eta$, without identifying an optimal face. Moreover x^ lies in the metric ball centered at x of radius $R_x = \sqrt{2\tilde{C}(x)/m_0}$. An inequality $A_i x \leq b_i$ is strictly inactive at x^* whenever $b_i - A_i x > R_x \|A_i\|_{H^{-1}}$. A switching sign is correct whenever $|(Dx - h)_a| > R_x \|D_a\|_{H^{-1}}$.*

Proof. For any feasible y , the tension-box inequality gives $-\lambda T(y) \leq -s^\top (Dy - h)$, while at x the difference is exactly $\Psi_s(x)$. Strong concavity therefore gives

$$W_\lambda(y) - W_\lambda(x) \leq (\nabla r(x) - D^\top s)^\top (y - x) - \frac{m_0}{2} \|y - x\|_H^2 + \Psi_s(x).$$

Insert e_x . Feasibility bounds the normal contributions by the first three terms in (40); completing the square proves (41). Strong convexity of $-W_\lambda$ and constrained optimality give $m_0 \|x - x^*\|_H^2/2 \leq W_\lambda(x^*) - W_\lambda(x)$. Cauchy–Schwarz in this ball proves the two strict tests. $\square$

The certificate is computable from a predicted actor and prices, including after exact feasibility repair. Its four leakage terms expose which continuation, boundary, or switching decision causes a first-order loss. For example, maximizing $-x - x^2/2$ on $[0, 1]$ loses $\epsilon + \epsilon^2/2$ at $x = \epsilon$: the lower-bound price contributes ϵ . Maximizing $-x^2/2 - |x|$ on $[-1, 1]$ gives the same loss at ϵ ; using $s = 0$ contributes switching leakage ϵ . Omitting either term invalidates the purported quadratic bound. Small norm error alone does not justify identifying a binding equality; the strict ball tests only discard inequalities or certify nonzero signs.

Weak duality and gap-based safe regions are established principles (Boyd and Vandenberghe, 2004; Fercoq et al., 2015). The relevant specialization here is an auditable loss for complete continuation policies, including *incorrect* predicted switching faces. It also prevents rounding a nominally fused edge from silently invalidating a subgradient certificate. Rational replay checks the repaired tiers, sign-independent tension capacities, and every leakage term. The bound can be a certificate-aware fitting objective, but its validity does not assume successful minimization of that objective.

6.5 Verifiable acceptance cells and compatible neural routing

For an implementable positive bridge, specialize the smooth resource to a quadratic and use the full coordinates of Proposition 4.5. Write the incremental problem as

$$\max_{Fy \leq d} \left\{ a(\theta)^\top y - \frac{1}{2} y^\top Q_B y - \lambda(\theta) \sum_n \kappa_n |(Ky)_n| \right\}, \quad Q_B = B^\top Q B \succ 0. \quad (42)$$

Here $Fy \leq d$ represents (23); the context enters a and $\lambda \geq 0$ affinely. The matrices and bounds are fixed. General nonlinear resources still have Theorem 6.4, but the exact affine representation below uses the quadratic assumption.

Choose a set I of binding inequalities, a set Z of fused edges, and signs $\sigma_n \in \{-1, 1\}$ on the remaining edges S . Set $C = [F_I^\top, K_Z^\top]^\top$ and $d_C = (d_I, 0)$. Require C to have full row rank, but do not require *all* binding constraints to be independent. Solve once for the affine maps in θ :

$$\begin{pmatrix} Q_B & C^\top \\ C & 0 \end{pmatrix} \begin{pmatrix} y_\theta \\ \mu_{I,\theta} \\ s_{Z,\theta} \end{pmatrix} = \begin{pmatrix} a(\theta) - K_S^\top (\lambda(\theta) \kappa_S \sigma_S) \\ d_C \end{pmatrix}. \quad (43)$$

Set inactive inequality prices to zero and $s_{S,\theta} = \lambda(\theta) \kappa_S \sigma_S$.

Theorem 6.5 (Verified cell, stability, and scalar critic). *Define the closed polyhedral parameter region*

$$C = \{\theta : \lambda(\theta) \geq 0, Fy_\theta \leq d, \mu_{I,\theta} \geq 0, |s_{Z,\theta}| \leq \lambda(\theta) \kappa_Z, \sigma_S \odot K_S y_\theta \geq 0\}. \quad (44)$$

At every $\theta \in C$, $z + By_\theta$ is the unique global accepted optimizer. No optimal multiplier at the query point is needed to check membership. If the nonconstant defining inequalities are $r_j(\theta) \geq 0$, then the region contains the parameter ball of radius

$$\rho(\theta) = \min_{j: \nabla r_j \neq 0} \frac{r_j(\theta)}{\|\nabla r_j\|_*} \quad (45)$$

around any of its points, where the norm is dual to the chosen parameter norm. Empty minima are interpreted as infinity. Boundary points can have radius zero.

Substitution in (42), with the fixed signs on S , gives a scalar quadratic value expert. The same construction with the full forcing p as parameter satisfies $\nabla_p J(p, \lambda) = z + By_\theta$ on its region. Along a lower-dimensional context map $p(\theta)$, its gradient is the corresponding chain-rule pullback, not an N -dimensional actor inferred

by inverting that map. Its friction derivative is $-T(x^*)$. Any learned rule that selects a cell and then verifies (44) returns this compatible value and policy wherever a cell is accepted. Different accepted cells agree on the policy and value at their overlap.

Proof. Positive definiteness of Q_B and full row rank of C make the saddle matrix nonsingular: a homogeneous solution has $Cy = 0$ and $y^\top Q_B y = 0$, hence $y = 0$, followed by a zero dual vector. Thus the maps in (43) are affine. Region membership supplies primal feasibility, dual nonnegativity, complementary slackness, and an absolute-value subgradient. The first block equation supplies stationarity. Convexity makes these conditions sufficient; strict convexity gives uniqueness. For each affine region inequality, perturbation by h changes its value by at most $\|\nabla r_j\|_* \|h\|$, proving (45). On the region, the absolute penalty uses fixed signs, so substitution yields a scalar quadratic. The envelope identity for a unique optimizer gives its forcing and friction derivatives. At a shared boundary uniqueness forces all verified representatives to give the same primal solution and objective. $\square$

A partially covered library is not claimed to supply a globally smooth approximate critic across rejection boundaries. The compatible gradient statement concerns its verified pieces; rejected queries use the separate policy certificate.

This is an offline–online construction, not an assertion that optimal faces can be learned without information. Offline optimizers generate candidate bases and affine price maps. A neural classifier learns to propose their indices. Online membership checks and Theorem 6.4 validate the proposed policy. A rejected or numerically inaccurate proposal invokes the specified static gate or a fully timed classical refinement. Unrepresented cells, degenerate bases, and incorrect classifier predictions cannot bypass the check. With fixed parameter dimension d_θ , an accepted query takes $O(Nd_\theta)$ arithmetic operations per proposed cell after its affine coefficients are stored. Offline factorization, library size, and bit complexity are not hidden in that bound.

Piecewise-affine control from multiparametric quadratic programming is classical (Bemporad et al., 2002), as are active-set path algorithms (Arnold and Tibshirani, 2016). Our bridge uses this machinery for *joint continuation and switching faces*, pairs each scalar expert with its complete acceptance-price map, and certifies arbitrary rejected-face error by (41). Together these results support a learn–verify–refine algorithm whose sufficient conditions contain no unknown query-time optimum. It does not assert full parameter-space coverage or neural superiority over classical cell lookup.

6.6 A distribution-free range for a declared parameter box

A useful validation range can be established from the model instead of the largest observed certificate. Let z be a fixed feasible outside policy with $T(z) = 0$, let forcing lie in a specified box, and let $\lambda \geq \lambda_{\min}$. Define incremental gain $G_\theta(x) = W_\theta(x) - W_\theta(z)$ and normalize by a fixed positive review mass S_0 . At every forcing-box vertex v , obtain a feasible policy x_v and a valid gap C_v at $\lambda_{\min}$.

Proposition 6.6 (Corner range and simultaneous gain validation). *Let $B = \max_v \{G_v(x_v) + C_v\}/S_0$. Every feasible policy gated against z has normalized gain in $[0, B]$, and regret at most B . For M frozen complete*

pipelines and n independent validation instances from their declared deployment distribution, set

$$h = B\sqrt{\log(2M/\delta)/(2n)}, \quad C_j = \min\{B, \tilde{C}_j/S_0\}.$$

With probability at least $1 - \delta$, simultaneously for every pipeline and for a validation-selected pipeline,

$$\mathbb{E} \text{Regret}_j/S_0 \leq \bar{C}_j + h, \quad \mathbb{E} G_j/S_0 \geq \max\{0, \bar{G}_j/S_0 - h\}. \quad (46)$$

Proof. For fixed z , optimal incremental gain is a maximum of affine functions of forcing and hence convex. Its maximum on a box is bounded by the largest vertex value. Increasing friction cannot increase incremental gain because $T(z) = 0$. Each vertex value is at most $G_v(x_v) + C_v$. Feasibility and the gate give the asserted ranges, including the range for regret. Apply one-sided Hoeffding inequalities to each certificate and each gain, then a union bound over $2M$ statements. Nonnegative gain supplies the zero lower bound. Simultaneity permits selection. $\square$

The concentration step remains standard. The new operational point is a deterministic range calculation and an actually frozen deployment rule. A parameter-dependent reoptimized outside policy need not preserve convexity of incremental gain; the corner argument is not applied to the predecessor noncentered experiment. Nor does a positive gain bound establish an advantage over a different nonstatic optimizer.

7 Nonsmooth service decisions and frozen learning pipelines

The experiments separate structural computation, candidate quality, and complete deployment. They do not use the generality of a theorem to imply empirical coverage of every admissible model. The earlier noncentered, boundary-inclusive and nonlinear studies remain intact in the companion; this section exercises nonzero absolute switching and the new verified-cell construction.

7.1 A declared service-control distribution and operational units

Consider binary scenario trees with discount factor 0.9 and equal branch probabilities. Thus $w_n = (0.9/2)^{\text{depth}(n)}$. Tiers lie in $[0, 1]$, the installed tier is $1/2$, and external net-payment coefficients are $c_n = w_n[0.8 + 0.1(n \bmod 5)]$, with nodes numbered from zero. The accepted polytope includes every non-root continuation cap and the root payment equality. Physical service and stock are held at their identified benchmark levels; the experiment concerns the remaining tier decision, not an untested inventory-learning claim.

The incremental profit, in profit units, is

$$\begin{aligned} G_{\alpha,\beta,\lambda}(x) = & \sum_n (g_{0n} + \alpha g_{1n} + \beta g_{2n})(x_n - \tfrac{1}{2}) - \sum_n w_n (x_n - \tfrac{1}{2})^2 \\ & - \tfrac{1}{10} \sum_{n \neq 0} w_n (x_n - x_{\text{pa}(n)})^2 - \lambda \sum_n w_n |x_n - x_{\text{pa}(n)}|, \end{aligned} \quad (47)$$

where the root’s predecessor is $1/2$ in the last term. Three fixed vectors describe net marginal resource offsets after the tariff; each has zero sum. Their exact rational coordinates and generating seed are recorded. Therefore the *reoptimized* fixed-tier comparator is exactly $1/2$ for every context: a common tier change has zero linear gain and strictly positive quadratic cost. This balanced-shock class is different from, and does not replace, the earlier noncentered class with boundary comparators.

One profit unit is the coefficient of a squared full-tier deviation in the per-review maintenance cost. The smooth inter-review cost has coefficient 0.1 , and λ prices an absolute unit of reconfiguration. Adding a common level cost can make all gross resource costs positive without changing decisions. Results are divided by the discounted review mass $S_0 = \sum_n w_n$. Multiplying all monetary primitives by a currency conversion multiplies every profit and certificate by that conversion; no such conversion is a field calibration. For example, a gain of 0.003 represents 0.30 per discounted review if one profit unit is defined as 100 currency units. No operational population is asserted to have that calibration.

7.2 Friction changes decisions, not only optimality labels

We solve 81 explicit epigraph quadratic programs: three contexts, trees with 15, 31, and 63 nodes, and nine friction values spanning zero to 1.25 times the computed critical value. Table 2 shows one context; every policy, tension, continuation price, and exact gap remains in the data. Nonzero friction changes both binding continuation constraints and fused edges. Switching decreases along the path and accepted gain disappears at the critical value to the certified tolerance. The zero-friction row is a control; the learning distribution below uses strictly positive friction.

The critical-friction study also includes binary and star trees from 31 to 1,023 nodes. Table 3 compares the factored and materialized linear programs using the same solver. On a star, forming *DB* creates quadratically many entries, while the two-factor representation retains $9N - 3$ nonzeros. Binary trees do not have this fill-in, and materialization can be faster there. The independent first-order algorithm produces rigorous brackets, but the 12,000-iteration cap does not always attain its 10^{-4} width target. Those failures remain visible. The result is a sparsity and certification consequence of the continuation structure, not a blanket speed claim for a custom optimizer.

7.3 Learning, verification, and actual refinement

The learning experiment uses 31-node problems. Context coordinates are independent uniforms on the rational grid $\{-1, -0.999, \dots, 1\}$; friction is independently uniform on $\{0.01, 0.011, \dots, 0.20\}$. We solve 1,024 training contexts, recover candidate bases, and retain 747 distinct verified cells. A 192-unit random-feature ReLU layer, with a fitted multiclass ridge readout, proposes cell indices. Its scalar experts and value gradients are given by Theorem 6.5. The classifier learns joint continuation and switching patterns; it is not the earlier derivative-weighted regressor, and no superiority of derivative supervision is inferred.

We freeze three static-gated policies: check the first neural proposal, check the first three neural proposals, or check the cell of the nearest training anchor. Each returns the first region-valid proposal after exact flow repair and certificate calculation, comparing its reward with static. If no candidate survives, it returns static.

Validation does not enlarge the library, update features, or tune a stopping rule. The complete model hash is recorded before and after both validation cohorts.

A separate 48-context matched-accuracy test adds actual classical refinement whenever the selected policy has normalized gap above 10^{-7} . It includes a cold epigraph quadratic program and sequential checking of the same frozen cell library. Table 4 includes generation, membership, exact audit, and all refinement time. The first-proposal face accuracy and added iteration counts distinguish learned candidate quality from a classical rescue. The data also contain raw face masks, switching signs, distances, and the acceptance and switching leakage of 96 neural/nearest proposals. Finite differences check the scalar experts' context-gradient pullbacks; these numerical checks supplement, not replace, the envelope proof.

Offline construction and fitting cost 18.22 seconds in the recorded environment. Under the observed mean timing difference, neural lookup recovers this cost after 4348 repeated queries relative to cold optimization; this is an accounting calculation, not a forecast or an advantage over sequential cell checking. The small, fixed-dimensional context distribution makes such reuse possible. The cell library can grow rapidly with problem complexity. Timings are single-environment descriptive measurements, and sequential checking is the stronger latency comparator in this run. The positive learning result is verified reuse with explicit rejection and acceptance guarantees, not a claim that the neural router dominates classical lookup.

7.4 Executed independent validation, including a useful gain certificate

A first cohort of 4,096 independent contexts uses the uniform range from the outside gradient residual. Its broad expectation bounds are retained in the data. We then fix a model-derived range using the four deterministic forcing corners at the minimum friction, without altering any policy. The confirmation plan is saved before a second, independent 4,096-context cohort is drawn. This prospective second cohort is conditional on the already frozen model and the declared range calculation; it is not a refit on the first cohort.

Table 5 executes Proposition 6.6 with $M = 3$, $n = 4096$, and $\delta = 0.05$. The selected three-cell neural policy has empirical gain 0.003006 and a simultaneous expected-gain lower bound of 0.001754 per discounted review relative to static. The normalized expected-regret upper bound is 0.036684. The gain guarantee is positive at the economic improvement scale, whereas the regret ceiling is too wide to establish near-optimality. Classical refinement, separately timed above, is needed for the high-accuracy deployment guarantee. No significant advantage over the nearest-anchor policy or over a fully optimized policy follows from this table.

7.5 Retained evidence and independent audit

The noncentered nonlinear study remains scientifically material: 48 instances, 432 deployments, boundary comparators, severe raw actor failures, and slower learned pipelines than cold SLSQP at matched accuracy. Its full text and tables are in Section EC.8; no unfavorable row is removed or pooled with the new distribution. The first four frozen predecessor seeds were an economical transfer diagnostic, not a representative sample of training variability. The inconclusive derivative-weighting comparison remains inconclusive. The earlier 992-coordinate failures and their actually executed fallbacks also remain available.

The new standard-library-only verifier independently reconstructs 10,312 rational policy certificates, 320,104 continuation checks, all corner ranges, and 12 rational critical-friction brackets. It imports neither the optimizer nor a numerical linear-algebra library. Optimizer flags, approximate face masks, and hardware timings are not treated as certificates. All data are synthetic; the mathematical scope, fixed-context validation scope, and previous transfer evidence are kept distinct.

8 Conclusions and operational implications

Accepted adaptation is a reallocation of real resources under commitments, not a transfer taken from the customer. On a tree, cumulative continuation prices and switching capacities determine which reallocations survive the exit option. The full-tree criterion extends the scalar edge test and distinguishes a zero-switching comparator’s critical friction from the interval of frictions supporting an already-changing outside protocol. Boundary and multiple-budget cases belong to the same balance system.

Learning enters through a verified value–decision representation rather than an assumed optimal face. Arbitrary predicted prices produce a full-horizon certificate with explicit continuation and switching leakage. Quadratic parameter cells make compatible experts usable without solving the query problem first; a learned index is accepted only after its region and policy are checked. Independent validation establishes a positive expected gain over static for a frozen policy in the declared distribution. It does not establish near-optimal raw learning, universal transfer, or superiority over classical cell lookup. The retained nonlinear failures and the new rejection rates identify where refinement remains necessary. Calibrating a service operation, changing tariffs endogenously, and treating private information are distinct extensions, not implicit features of the present model.

A Primitive structure under continuation participation

Continuation participation changes which customer-neutral reallocations are permissible. We give a complete scalar tree criterion and a quantitative improvement that can be checked from local primitives, without solving the defining nonlinear control program. The graph formulation above remains available for portfolios; the scalar result is not silently promoted to a graph theorem.

A.1 An edge test rather than a high-dimensional direction search

Let o be the root of a finite decision tree. Assume $a_n > 0$, $w_n > 0$, scalar continuous tiers, and an interior constant comparator $\bar{\theta}$. Define $c_n = w_n a_n$ and the accepted, payment-neutral class

$$\sum_n c_n (x_n - \bar{\theta}) = 0, \quad \sum_{j \succeq n} c_j (x_j - \bar{\theta}) \leq 0 \quad (n \neq o), \quad 0 \leq x_n \leq 1. \quad (48)$$

Here $j \succeq n$ includes n itself. Root equality holds customer utility exactly fixed; every nonroot inequality prevents the remaining expected payment from exceeding the static outside protocol. Identified fill and

physical cost stay fixed. This is a subclass of weak acceptance, so every strict improvement obtained here also improves the weakly accepted problem.

Let $\mathcal{R}$ be convex and differentiable, and define the primitive marginal reward per payment unit at the comparator by

$$k_n = \frac{w_n a_n - \partial_n \mathcal{R}(\bar{\theta} \mathbf{1})}{w_n a_n}. \quad (49)$$

For scalar maintenance and centered smooth switching, $A'_n(0) = 0$ and $\nabla \mathcal{G}(\bar{\theta} \mathbf{1}) = 0$, this is $k_n = 1 - M'_n(\bar{\theta})/a_n$ at every nonroot node. At the root, subtract also $A'_o(\bar{\theta} - q_0)/a_o$. Thus the test permits nonquadratic maintenance, arbitrary finite smooth friction, and nonstationary tariffs. Initial installation is not omitted.

Theorem A.1 (Complete continuation-compatible edge criterion). *In (48), the static protocol is globally optimal if and only if*

$$k_n \geq k_{\text{pa}(n)} \quad \text{on every nonroot edge.} \quad (50)$$

If some edge $u \rightarrow v$ has $\Delta = k_u - k_v > 0$, set $d_u = 1/c_u$, $d_v = -1/c_v$, and all other components to zero. Then $\bar{\theta} \mathbf{1} + sd$ is feasible for

$$0 \leq s \leq s_{\max} = \min\{c_u(1 - \bar{\theta}), c_v \bar{\theta}\},$$

and strictly improves the reward for all sufficiently small positive s . If $d^\top \nabla^2 \mathcal{R}(\bar{\theta} \mathbf{1} + sd)d \leq L_d$, its gain is at least $s\Delta - L_d s^2/2$. The edge test costs $O(N)$ comparisons once the primitive marginal costs have been evaluated.

Proof. For any feasible x , put $y_n = c_n(x_n - \bar{\theta})$ and $S_n = \sum_{j \succeq n} y_j$. Then $S_o = 0$ and $S_n \leq 0$ for $n \neq o$. Since $y_n = S_n - \sum_{v: \text{pa}(v)=n} S_v$, finite summation by parts gives

$$\nabla W(\bar{\theta} \mathbf{1})^\top (x - \bar{\theta} \mathbf{1}) = \sum_{n \neq o} (k_n - k_{\text{pa}(n)}) S_n. \quad (51)$$

Condition (50) makes this nonpositive for every feasible x ; concavity proves global optimality. If an edge violates it, the displayed two-coordinate transfer has $S_v = -s$ and all other nonroot subtree sums zero. It therefore satisfies every continuation inequality, not just an ex ante budget. Its derivative is $\Delta > 0$, and the box gives $s_{\max}$. The curvature bound proves the finite gain. This also proves necessity without appealing to existence of unknown optimizer multipliers. $\square$

The direction moves expected payment from a descendant review to its parent. This timing respects a customer's later option to leave. By contrast, shifting payment from a low-price state to a high-price sibling can violate that sibling's continuation promise, even when total payment is unchanged. The edge criterion exposes precisely the order imposed by this institution. For weak root acceptance instead of equality, the same calculation adds $k_o S_o$; the complete scalar criterion is (50) together with $k_o \geq 0$.

A primitive curvature bound is obtained by summing all affected terms:

$$L_d \leq \sum_n w_n \bar{M}_n d_n^2 + \sum_n w_n \bar{A}_n (d_n - d_{\text{pa}(n)})^2 + \bar{G}_d, \quad d_{\text{pa}(o)} = 0. \quad (52)$$

Here $\overline{M}_n, \overline{A}_n$ bound second derivatives along the entire feasible segment, and $\overline{G}_d$ bounds the joint coupling curvature. The sum includes incoming and outgoing switching edges, including unchanged children of either perturbed node. If $L_d = L_0 + \lambda L_A$, an untruncated step yields a guaranteed gain $\Delta^2/[2(L_0 + \lambda L_A)]$. Finite smooth adjustment can make the gain small without necessarily eliminating it.

A.2 An exact threshold for nonsmooth switching

The contrast with absolute adjustment is substantive rather than notational. Consider a two-review chain $o \rightarrow v$, $q_0 = \bar{\theta}$, the same root payment equality and child participation constraint, and switching cost $\lambda\{\kappa_o|x_o - q_0| + \kappa_v|x_v - x_o|\}$, where $\kappa_o, \kappa_v > 0$ include their probability weights. Let k_o, k_v now exclude that absolute switching term, and let the remaining resource cost be differentiable and convex. Set

$$K = \kappa_o/c_o + \kappa_v(1/c_o + 1/c_v), \quad \lambda_c = \max\{0, (k_o - k_v)/K\}.$$

Corollary A.2 (Friction threshold with an exit option). *The static protocol is globally optimal exactly when $\lambda \geq \lambda_c$. For $0 \leq \lambda < \lambda_c$, a sufficiently small customer-neutral transfer from the child to its parent strictly improves reward while satisfying continuation participation.*

Proof. Every feasible displacement has $d_o = s/c_o$, $d_v = -s/c_v$ for $s \geq 0$. Its right derivative at zero is $k_o - k_v - \lambda K$. A positive derivative gives improvement. A nonpositive derivative, combined with concavity on this one-dimensional feasible interval, rules out improvement everywhere, including the equality case. The threshold therefore uses primitive rates and frictions, not an uncomputed optimizer. $\square$

A.3 What the value of contractual memory does and does not mean

Proposition A.3 (Zero-friction bound without history-specific promises). *With exogenous regimes, compact convex tiers in $[0, 1]^d$, convex stage resources, and only ex ante linear payment commitments, assume that stage resource functions and payment coefficients depend only on time and the current regime, with no other history-dependent feasibility or physical carry-over. Let $W_{\text{full}}(\lambda)$ and $W_{t,z}(\lambda)$ allow full observed history or only time and current regime. Suppose the sole inherited-tier term is $\lambda \mathbb{E} \sum_t \beta^t \|x_t - x_{t-1}\|^2$. For any time-regime optimizer π_0 at zero friction,*

$$0 \leq W_{\text{full}}(\lambda) - W_{t,z}(\lambda) \leq \lambda \mathcal{A}(\pi_0) \leq \lambda d \sum_t \beta^t.$$

Proof. At zero friction, conditional expectation of actions given (t, z_t) preserves expected linear commitments and reduces convex stage costs. Thus both zero-friction values coincide and compactness gives π_0 . Adding nonnegative friction reduces the full value, whereas using π_0 loses at most $\lambda \mathcal{A}(\pi_0)$. Compact tier bounds give the last inequality. $\square$

The proposition deliberately excludes history-specific promises. This exclusion has content: with two equiprobable histories leading to the same terminal regime, terminal utility $x - x^2$, and respective tier caps $1/4$ and $3/4$, history-dependent control earns $(3/16 + 1/4)/2 = 7/32$, whereas a common terminal tier earns

3/16. The gap is $1/32$ even at zero friction. Both histories and their caps are public. Different earlier installed tiers can record which promise applies. The missing information concerns a contractual obligation, not a new signal about demand. Consequently no unconditional claim that continuation-constrained memory value vanishes is made.

For the continuation institution, a remaining-payment budget supplies the appropriate dynamic state. At node n , with incoming budget b and outside-protocol cap $\bar{b}_n$, choose the tier x and child budgets b_v subject to

$$a_n^\top x + \beta \sum_{v \in \text{ch}(n)} P_{nv} b_v \leq \min\{b, \bar{b}_n\}. \quad (53)$$

The Bellman state is (n, q, b) , or (t, z, q, b) when the subtree primitives and outside caps are Markov. Its continuation term is $\beta \sum_v P_{nv} V_v(x, b_v)$. Backward induction establishes equivalence with the full tree program: feasible child budgets give a feasible subtree protocol, while a feasible protocol supplies child budgets equal to its actual expected remaining payments. At zero friction the recursion does not use q , but it still uses the promise budget and the information determining the outside caps. This is the precise continuation-aware replacement for conditioning on demand regime alone.

Data, software, and computation accessibility

The versioned review package contains ordinary manuscript inputs, complete proofs, frozen training coefficients, generated test primitives, rational policies and multipliers, independent certificate replay, raw execution times, and source hashes. The README identifies the current build and validation entry points. Predecessor source files and scientific observations are preserved, with an explicit relocation map. All experiments are synthetic; no confidential or personal data are used.

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Table 1: Accepted continuous-tier information values

Information	Provider reward	Increment	Payment price
Static	-5.870110973	—	0.000000000
Time	-5.869294096	0.000816877	0.001568294
Time–regime	-5.252736080	0.616558016	0.174849751
Full state	-5.233445701	0.019290379	0.180271244

All four classes have the same filled demand 18.021386714, customer utility 28.657506145, and physical cost 9.587377732. All policies are deterministic. Decimals round exact rational values; stock is optimized over the full interval through Lemma 5.1.

Table 2: Accepted decisions along nonzero switching-friction paths

Nodes	λ/λ_c	λ	Gain	Switching	Binding caps	Fused edges
15	0.00	0.0000	0.008837	0.08549	8	0
15	0.50	0.2066	0.001294	0.01253	13	12
15	1.00	0.4131	0.000000	0.00000	14	15
15	1.25	0.5164	0.000000	0.00000	14	15
31	0.00	0.0000	0.014286	0.09843	14	0
31	0.50	0.2347	0.002270	0.02281	22	15
31	1.00	0.4694	0.000000	0.00000	30	31
31	1.25	0.5867	0.000000	0.00000	30	31
63	0.00	0.0000	0.015029	0.11327	26	2
63	0.50	0.2823	0.001509	0.01341	45	41
63	1.00	0.5646	0.000000	0.00000	62	63
63	1.25	0.7058	0.000000	0.00000	62	63

Context (0.2, -0.4); all 81 paths across three contexts are retained in the data. Gain and switching are per discounted review. Binding and fused masks use 10^{-6} for numerical diagnosis; exact rational policy gaps, rather than masks, determine validity. The zero-friction row is the control, not the learning experiment.

Table 3: Factored critical-friction computation and independent brackets

Tree	Nodes	Factored entries	Expanded entries	Factored ms	Expanded ms	Bracket width
Binary	31	276	242	2.26	2.56	0.00003
Binary	127	1,140	1,010	3.83	3.74	0.00009
Binary	1023	9,204	8,178	33.63	23.97	0.00674
Star	31	276	1,054	1.55	2.63	0.00008
Star	127	1,140	16,510	3.43	7.03	0.00008
Star	1023	9,204	1,049,598	51.70	462.99	0.00175

Both linear programs use HiGHS; expanded time includes materializing DB . The final column is the exact rational width from the separate accelerated projected-gradient algorithm, capped at 12,000 iterations and targeting 10^{-4} . Widths above target are retained, not recorded as successful solves. Sparse arithmetic is not a claim that this first-order algorithm beats the linear-programming solver.

Table 4: Complete deployment at a common certified gap

Method	Milliseconds	Immediate (%)	Refine (%)	First face (%)	Added iterations
Cold quadratic program	19.74	–	–	–	–
Neural, three cells	17.94	31.2	68.8	16.7	-7.90
Nearest anchor	19.82	20.8	79.2	22.9	-5.21
Sequential cell check	5.83	52.1	47.9	0.0	-13.27

Forty-eight fresh 31-node problems. Time includes proposal generation, membership checking, exact rational audit, and any executed refinement. Refused cells use a cold classical refinement, not an unreported warm start. All final normalized gaps are at most 10^{-7} . First-face masks are descriptive and use the tolerance stated in the data. Offline construction is reported separately; these timings are hardware-dependent observations, not population speed claims.

Table 5: Independent validation of frozen, static-gated pipelines

Pipeline	Cell accepted (%)	Mean gain	Expected gain floor	Expected regret ceiling
Neural, one cell	16.43	0.001734	0.000482	0.042906
Neural, three cells	28.86	0.003006	0.001754	0.036684
Nearest anchor	21.34	0.002291	0.001039	0.040445

A separate sample of $n = 4096$ contexts, three frozen pipelines, and $\delta = 0.05$. Both expectation bounds hold simultaneously under the declared deployment distribution. The model-derived range is $B = 0.051784$ and the deviation radius is 0.001252. The gain floor compares with static, not with an optimized nonstatic policy. The regret ceilings are valid but not small enough to imply near-optimality.